

Multimodal Foundations

DATA 443/643 · Week 4, note 4.1

Week 4

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Ideas from CMU 11-777 (five jobs; joint vs coordinated). Original slides; that course is not copied.

Lecture plan

- 1 Why more than one stream
- 2 Five jobs: represent, align, fuse, translate, co-learn
- 3 Joint concat versus two towers and a cosine
- 4 A 2-D retrieval numeric

Key goal: leave knowing CLIP scores pairs; it does not fuse pixels with token ids.

Five jobs, not one architecture

- Represent: one vector, or two towers?
- Align, fuse, translate, co-learn are different verbs.
- Captioning maps. CLIP scores.

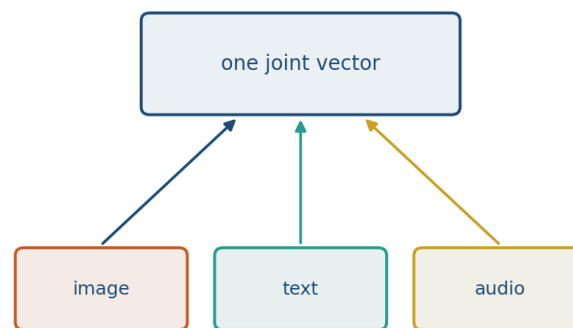
Five jobs. CLIP scores; captioning translates; concat fuses.



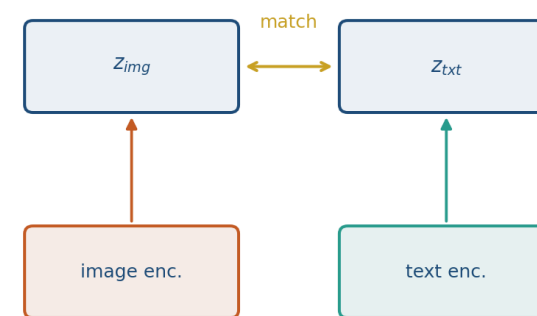
Joint versus coordinated

- Joint: mash streams into one vector.
- Coordinated: a tower each, then a similarity.
- Missing a stream hurts the joint model.

Joint: fuse into one space



Coordinated: keep towers, pull them together



What your demo is allowed to do

Typed search over photos

Coordinated. Embed the query alone. Rank by cosine.

Must output a sentence

Translation. You need a decoder (BLIP, an LLM). Cosine is not a caption.

Rank by cosine

- Match $v = [1, 0]$, $t = [0.8, 0.2]$ is about 0.97.
- Mismatch with $[0, 1]$ is 0.
- Concat of those two vectors is 4-D and needs both.

Coordinated retrieval ranks by cosine. Joint concat needs both streams.

pair	vectors	cosine
match	$v = [1, 0]$, $t = [0.8, 0.2]$	0.97
mismatch	$v = [1, 0]$, $t = [0, 1]$	0

What to take from this lecture

Do not concat and call it CLIP

CLIP never concatenates pixels with token ids in pretraining.

Name the job

Search, caption, or detect a region?
Different boxes on the five-job strip.

Week 4.3

The coordination loss is InfoNCE.
Zero-shot is a text prompt.